

1-5

Support Reactions: For member AB

$$\zeta + \Sigma M_B = 0; \quad 9.00(4) - A_y(12) = 0 \quad A_y = 3.00 \text{ kip}$$

$$\rightarrow \Sigma F_x = 0; \quad B_x = 0$$

$$+\uparrow \Sigma F_y = 0; \quad B_y + 3.00 - 9.00 = 0 \quad B_y = 6.00 \text{ kip}$$

Equations of Equilibrium: For point D

$$\rightarrow \Sigma F_x = 0; \quad N_D = 0$$

$$+\uparrow \Sigma F_y = 0; \quad 3.00 - 2.25 - V_D = 0$$

$$V_D = 0.750 \text{ kip}$$

$$\zeta + \Sigma M_D = 0; \quad M_D + 2.25(2) - 3.00(6) = 0$$

$$M_D = 13.5 \text{ kip} \cdot \text{ft}$$

Equations of Equilibrium: For point E

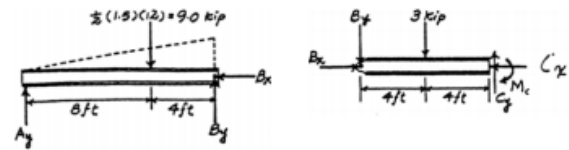
$$\rightarrow \Sigma F_x = 0; \quad N_E = 0$$

$$+\uparrow \Sigma F_y = 0; \quad -6.00 - 3 - V_E = 0$$

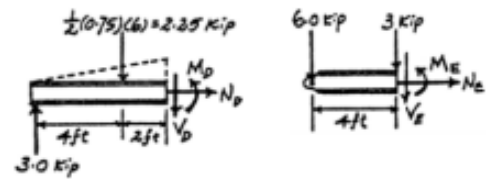
$$V_E = -9.00 \text{ kip}$$

$$\zeta + \Sigma M_E = 0; \quad M_E + 6.00(4) = 0$$

$$M_E = -24.0 \text{ kip} \cdot \text{ft}$$



Ans.



Ans.

Negative signs indicate that M_E and V_E act in the opposite direction to that shown on FBD.

1-71. Determine the average normal stress at section $a-a$ and the average shear stress at section $b-b$ in member AB . The cross section is square, 0.5 in. on each side.

Consider the FBD of member BC , Fig. a ,

$$\zeta + \sum M_C = 0; \quad F_{AB} \sin 60^\circ(4) - 150(4)(2) = 0 \quad F_{AB} = 346.41 \text{ lb}$$

Referring to the FBD in Fig. b ,

$$\nearrow \sum F_x = 0; \quad N_{a-a} + 346.41 = 0 \quad N_{a-a} = -346.41 \text{ lb}$$

Referring to the FBD in Fig. c ,

$$+\uparrow \sum F_y = 0; \quad V_{b-b} - 346.41 \sin 60^\circ = 0 \quad V_{b-b} = 300 \text{ lb}$$

The cross-sectional areas of section $a-a$ and $b-b$ are $A_{a-a} = 0.5(0.5) = 0.25 \text{ in}^2$ and

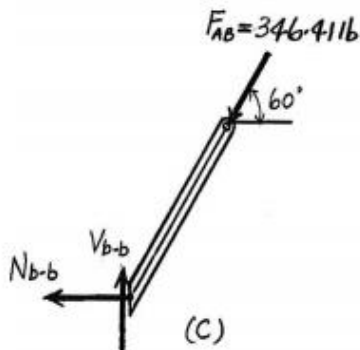
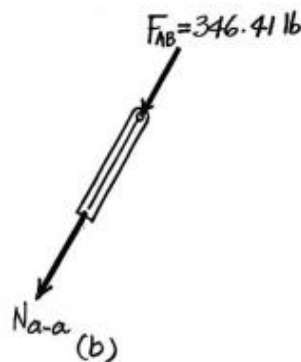
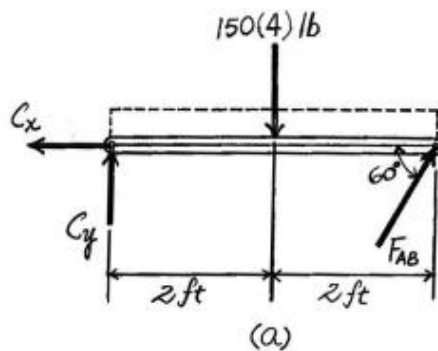
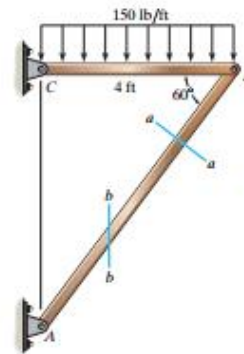
$$A_{b-b} = 0.5 \left(\frac{0.5}{\cos 60^\circ} \right) = 0.5 \text{ in}^2. \text{ Thus}$$

$$\sigma_{a-a} = \frac{N_{a-a}}{A_{a-a}} = \frac{-346.41}{0.25} = -1385.64 \text{ psi} = 1.39 \text{ ksi}$$

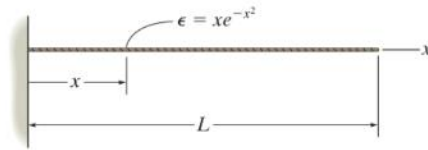
Ans.

$$\tau_{b-b} = \frac{V_{b-b}}{A_{b-b}} = \frac{300}{0.5} = 600 \text{ psi}$$

Ans.



- 2-28 *2-28. The wire is subjected to a normal strain that is defined by $\epsilon = xe^{-x^2}$, where x is in millimeters. If the wire has an initial length L , determine the increase in its length.

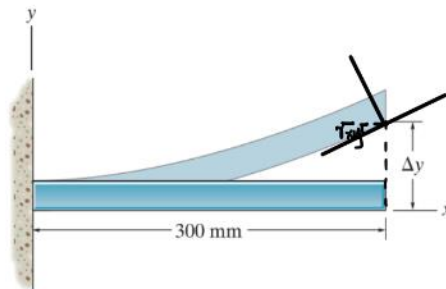


Prob. 2-28

$$\delta L = \epsilon dx = xe^{-x^2} dx$$

$$\therefore \Delta L = \int_0^L xe^{-x^2} dx = -\frac{e^{-x^2}}{2} \Big|_0^L = -\frac{e^{-L^2}}{2} + \frac{1}{2} = -\frac{1}{2}(e^{-L^2} - 1)$$

- 2-32 *2-32. The bar is originally 300 mm long when it is flat. If it is subjected to a shear strain defined by $\gamma_{xy} = 0.02x$, where x is in meters, determine the displacement Δy at the end of its bottom edge. It is distorted into the shape shown, where no elongation of the bar occurs in the x direction.



Prob. 2-32

$$\frac{dy}{dx} = \tan \gamma_{xy} = \tan(0.02x)$$

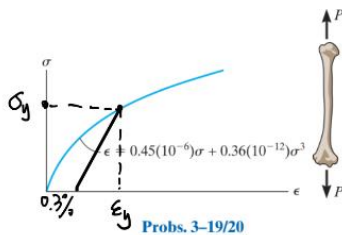
$$dy = \tan(0.02x) dx$$

$$\therefore \Delta y = \int_0^{300 \text{ mm}} dy = \int_0^{300 \text{ mm}} \tan(0.02x) dx$$

$$= \left| \frac{1}{0.02} \ln \cos(0.02x) \right|_0^{300 \text{ mm}}$$

$$= 2.03 \text{ mm}$$

- 3-19. The stress-strain diagram for a bone is shown, and can be described by the equation $\epsilon = 0.45(10^{-6})\sigma + 0.36(10^{-12})\sigma^3$, where σ is in kPa. Determine the yield strength assuming a 0.3% offset.

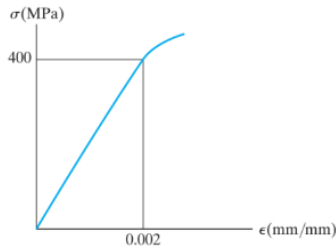


Probs. 3-19/20

$$E = \left. \frac{d\sigma}{d\epsilon} \right|_{\sigma=0} = \frac{1}{0.45 \times 10^{-6}}$$

$$\begin{cases} \frac{\sigma_y}{\epsilon_y - 0.3\%} = E \\ \epsilon_y = 0.45 \times 10^{-6} \sigma_y + 0.36 \times 10^{-12} \sigma_y^3 \\ \sigma_y > 0 \\ \epsilon_y > 0 \end{cases} \Rightarrow \sigma_y = 2027 \text{ kPa}$$

3-27. The elastic portion of the stress-strain diagram for a steel alloy is shown in the figure. The specimen from which it was obtained had an original diameter of 13 mm and a gauge length of 50 mm. When the applied load on the specimen is 50 kN, the diameter is 12.99265 mm. Determine Poisson's ratio for the material.



Prob. 3-27

diameter $\rightarrow$ x axis
length $\rightarrow$ y axis

$$\sigma = \frac{P}{A} = \frac{50 \text{ kN}}{\pi \left(\frac{13}{2}\right)^2 \text{ mm}^2} = 376.7 \text{ MPa}$$

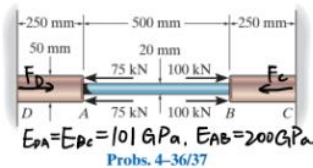
$$\epsilon_x = \frac{376.7}{400} \times 0.002 \text{ mm/mm} = 0.00188 \text{ mm/mm}$$

$$\epsilon_y = \frac{12.99265 - 13}{13} = -0.565 \times 10^{-3} \text{ mm/mm}$$

$$\nu = -\frac{\epsilon_y}{\epsilon_x} = 0.30$$

*4-36. The composite bar consists of a 20-mm-diameter A-36 steel segment AB and 50-mm-diameter red brass C83400 end segments DA and CB. Determine the average normal stress in each segment due to the applied load.

*4-37. The composite bar consists of a 20-mm-diameter A-36 steel segment AB and 50-mm-diameter red brass C83400 end segments DA and CB. Determine the displacement of A with respect to B due to the applied load.



Probs. 4-36/37

$$4-36: A_{AB} = \pi \left(\frac{20}{2}\right)^2 \text{ mm}^2 = 0.00314 \text{ m}^2$$

$$A_{DA} = A_{BC} = \pi \left(\frac{50}{2}\right)^2 \text{ mm}^2 = 0.00196 \text{ m}^2$$

$$\leftarrow \sum F_x = 0: F_C - F_D + 2 \times 75 \text{ kN} - 2 \times 100 \text{ kN} = 0$$

$$\Rightarrow F_C - F_D = 50 \text{ kN}$$

$$\leftarrow \frac{F_D L_{AD}}{A_{DA} E_{DA}} - \frac{F_C L_{BC}}{A_{BC} E_{BC}} + \frac{(2 \times 75 \text{ kN} - F_D) L_{AB}}{A_{AB} E_{AB}} = 0$$

$$\Rightarrow F_D = 107.89 \text{ kN}, F_C = 157.89 \text{ kN}$$

$$\sigma_{DA} = \frac{P_{DA}}{A_{DA}} = \frac{107.89 \text{ kN}}{0.00196 \text{ m}^2} = 55.0 \text{ MPa}$$

$$\sigma_{BC} = \frac{P_{BC}}{A_{BC}} = \frac{157.89 \text{ kN}}{0.00196 \text{ m}^2} = 80.4 \text{ MPa}$$

$$\sigma_{AB} = \frac{P_{AB}}{A_{AB}} = \frac{(150 - 107.89) \text{ kN}}{0.00314 \text{ m}^2} = 13.4 \text{ MPa}$$

*4-68. A steel surveyor's tape is to be used to measure the length of a line. The tape has a rectangular cross section of 0.05 in. by 0.2 in. and a length of 100 ft when $T_1 = 60^\circ\text{F}$ and the tension or pull on the tape is 20 lb. Determine the true length of the line if the tape shows the reading to be 463.25 ft when used with a pull of 35 lb at $T_2 = 90^\circ\text{F}$. The ground on which it is placed is flat. $\alpha_{st} = 9.60(10^{-6})/^\circ\text{F}$, $E_{st} = 29(10^3) \text{ ksi}$.



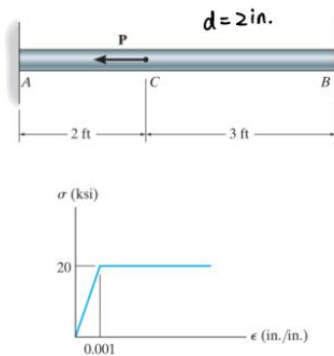
Prob. 4-68

$$\delta_T = \alpha \Delta T L = 9.6 \times 10^{-6} \times (90 - 60) \times 463.25 = 0.133 \text{ ft}$$

$$\delta = \frac{PL}{AE} = \frac{(35 - 20) \times 463.25}{0.2 \times 0.05 \times 29 \times 10^6} = 0.024 \text{ ft}$$

$$L = 463.25 + 0.133 + 0.024 = 463.407 \text{ ft}$$

*4-112. Determine the elongation of the bar in Prob. 4-111 when both the load P and the supports are removed.



Probs. 4-111/112

$$E = \frac{20 \text{ ksi}}{0.001 \text{ in./in.}} = 2 \times 10^4 \text{ ksi.}$$

$$F_A = F_B = \sigma A = 20 \pi (1)^2 = 62.832 \text{ kip}$$

$$\rightarrow \sum F_x = 0 \quad F_A + F_B - P = 0 \Rightarrow P = 126 \text{ kip}$$

$$\delta_C = \epsilon L = (0.001) \times 3 \times 12 = 0.036 \text{ in.}$$

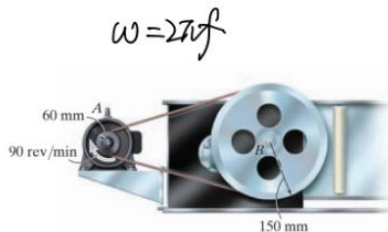
$$\frac{F_A' \cdot 2}{AE} = \frac{F_B' \cdot 3}{AE} \Rightarrow F_A' = 1.5 F_B'$$

$$\Rightarrow F_B' = 0.4P, F_A' = 0.6P$$

$$F_A'' = F_B'' = -62.832 + 0.6 \times 125.66 = 62.832 - 0.4 \times 125.66 = 12.568 \text{ kip.}$$

$$\delta = \frac{PL}{AE} = 0.0120 \text{ in.}$$

5-38. The motor A develops a power of 300 W and turns its connected pulley at 90 rev/min. Determine the required diameters of the steel shafts on the pulleys at A and B if the allowable shear stress is $\tau_{\text{allow}} = 85 \text{ MPa}$.



Prob. 5-38

$$T_A = \frac{P}{2\pi f_A} = \frac{300}{2\pi \times \frac{90}{60}} = \frac{100}{\pi} \text{ N}\cdot\text{m} \approx 31.8 \text{ N}\cdot\text{m}$$

$$W_A \cdot r_A = W_B \cdot r_B \Rightarrow f_B = \frac{W_B}{2\pi} = \frac{3}{2\pi} \text{ rev/s} \Rightarrow T_B = \frac{P}{2\pi f_B} \approx 79.58 \text{ N}\cdot\text{m}$$

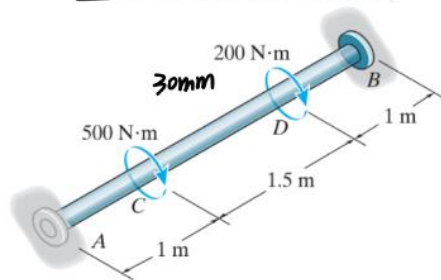
$$\tau_{\text{allow}} = \frac{T_C}{J} = \frac{2T}{\pi c^3} = 85 \text{ MPa}$$

$$\therefore c_A = 6.2 \text{ mm}$$

$$c_B = 8.4 \text{ mm}$$

$$\therefore d_A = 12.4 \text{ mm}, d_B = 16.8 \text{ mm}$$

5-78. The A-36 steel shaft has a diameter of 60 mm and is fixed at its ends A and B . If it is subjected to the torques shown, determine the absolute maximum shear stress in the shaft.



Prob. 5-78

$$\sum M_x = 0: T_A + T_B - 500 \text{ N}\cdot\text{m} - 200 \text{ N}\cdot\text{m} = 0.$$

At the end A :

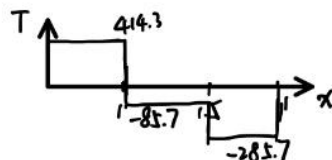
$$\phi_{TA} - \phi_T = 0$$

$$\therefore \frac{T_A \times 3.5}{JG} - \frac{500 \times 1.5}{JG} - \frac{700 \times 1}{JG} = 0$$

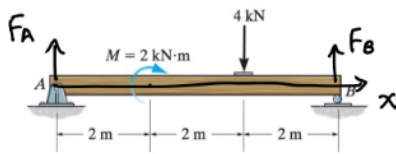
$$\therefore T_A = 414.3 \text{ N}\cdot\text{m}$$

$$\therefore T_B = 285.7 \text{ N}\cdot\text{m}$$

$$\tau_{\text{max}} = \frac{T_{AC} c}{J} = 9.77 \text{ MPa.}$$



6-2. Draw the shear and moment diagrams for the simply supported beam.



Prob. 6-2

Support reactions.

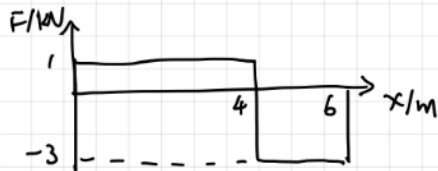
$$F_A + F_B = 4 \text{ kN.}$$

$$F_B \times 6 \text{ m} = 4 \text{ kN} \times 4 \text{ m} + M \Rightarrow F_A = 1 \text{ kN}$$

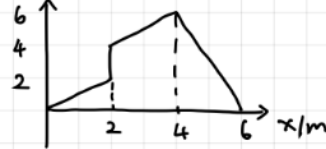
$$F_B = 3 \text{ kN}$$

$$F_A \times 6 \text{ m} + M = 4 \text{ kN} \times 2 \text{ m}$$

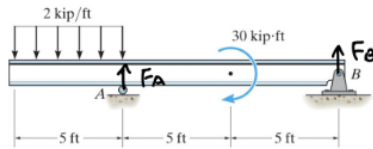
Shear:



Moment:



6.19 6-19. Draw the shear and moment diagrams for the beam.



Prob. 6-19

$$F_A + F_B = 10 \text{ kip}$$

$$10 \text{ kip} \times 2.5 \text{ ft} + F_B \times 10 \text{ ft} = 30 \text{ kip} \cdot \text{ft}$$

$$\Rightarrow F_A = 9.5 \text{ kip}, F_B = 0.5 \text{ kip}$$

Shear:



Moment:

